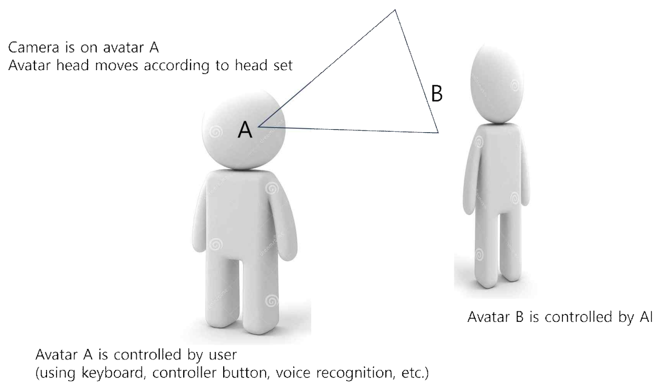
Introduction to Virtual Reality, 2026 Spring

Assignment 4: Project Refinement (due end of semester)

1. "Where" cues to improve spatial presence – Refine your VR flirting scene with objects that strengthen the sense of "place" and "space". Examples can include floors, walls, landmarks attached to the floors, space related objects (e.g. if the place is outdoor in spring, put up a cherry blossom tree), familiar objects whose size can be used to judge absolute distances in the scene (e.g. smartphone, coke). Include/arrange objects that take advantage of the psychological 3D effects (occlusion, motion parallax, haziness, etc.)

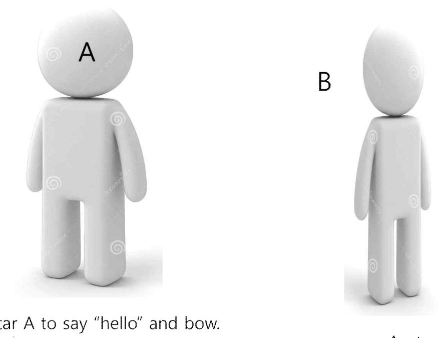
   Augmented reality – Using the markers included in the 360 degree video backdrop, overlay objects to decorate your VR flirting scene.

2. Extend Homework 3 – Situate two human characters (A and B) in the scene, and have them sit or stand facing each other. Make it so that A is controlled by you and as a user (you) can share Avatar A's view point. You should be able to control Avatar A in simple way for various behaviors (keyboard, interaction controller/button command, or even voice command) – e.g. stand, sit, wave hand, bow, say a word, smile, frown, …



   Make the other human figure B to be controlled by AI (see below).

3. AI feature and affective quality – Build a small AI engine (using open source LLM and simple network communication) to make simple conversation and implement animated control scheme for the Avatar B. See the picture below. Try to make Avatar B to have a particular persona (e.g. impressionable early 20's) so that the responses can convey some persona specific or emotional qualities. Experiment of the VR UX when the interaction is simplistic (e.g. no gestures (just voice), no persona, …).



You control Avartar A to say "hello" and bow.

Avatar B says "Hi" and, bows and smiles.

LLM gets a prompt: "Voice: Hello, Action: Bow"

LLM produces response: "Voice: Hi, Action: Bow, Smile"

Augmented 3D objects (placards, photo zone) / Billboard objects
Particles and special effects
Portals and Video façade
Stereo / Shadow
Self body features

Multi-user (2)
Gesture recognition / Voice recognition / Prose analysis
Avatar tracking and animation

Interact with objects (take photos)
Multimodal – gestures, 2D selection, voice, …
Navigation and sickness reduction
Markertracking / Device interfacing

Positive / Negative
 Expression, hair style, clothing, Place, …

Live video demo and oral report and submit to KULMS